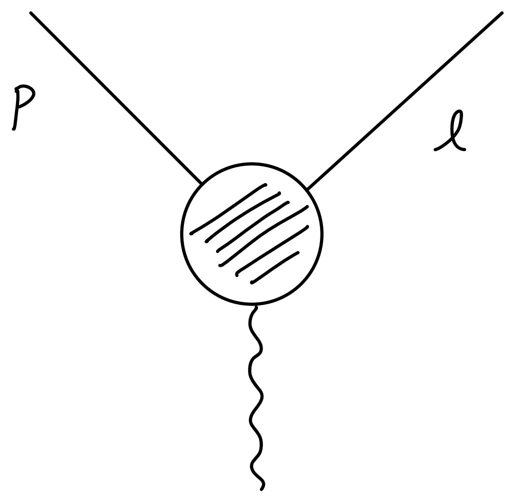
Prelude: SCET



$$Q^2 = (p-\ell)^2$$

$$\lambda^2 \sim \frac{p^2}{Q^2} \sim \frac{\ell^2}{Q^2}$$

$$n^\mu = (1, 0, 0, 1)$$

$$\bar{n}^\mu = (1, 0, 0, -1)$$

$$p^\mu = (n \cdot p) \frac{\bar{n}^\mu}{2} + (\bar{n} \cdot p) \frac{n^\mu}{2} + p_\perp^\mu$$

$$= (p_+, p_-, p_\perp^\mu)$$

$$\rightarrow \quad p^\mu \sim (\lambda^2, 1, \lambda)\, Q \quad , \quad \ell^\mu \sim (1, \lambda^2, \lambda)\, Q$$

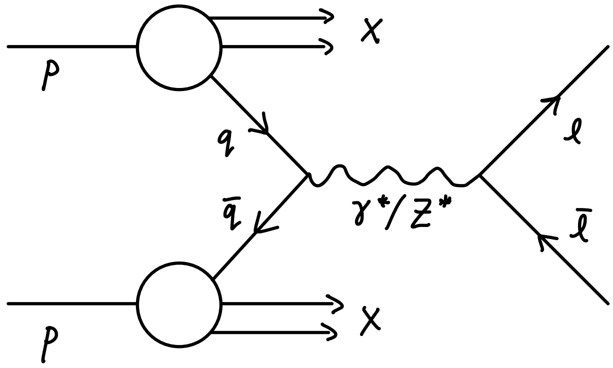
$$\sum_{i=0}^{\infty} \frac{t^i}{i!} (\bar{n} \cdot \partial)^i \varphi_c(x) = \varphi_c(x + t\bar{n})$$

$$\bar{\chi}(x) = \bar{\Psi}(x) \, P \exp\left[ig \int_{-\infty}^{0} ds \; \bar{n} \cdot A(x + s\bar{n})\right]$$

TMD Handbook Ch 2: Definition of TMDs

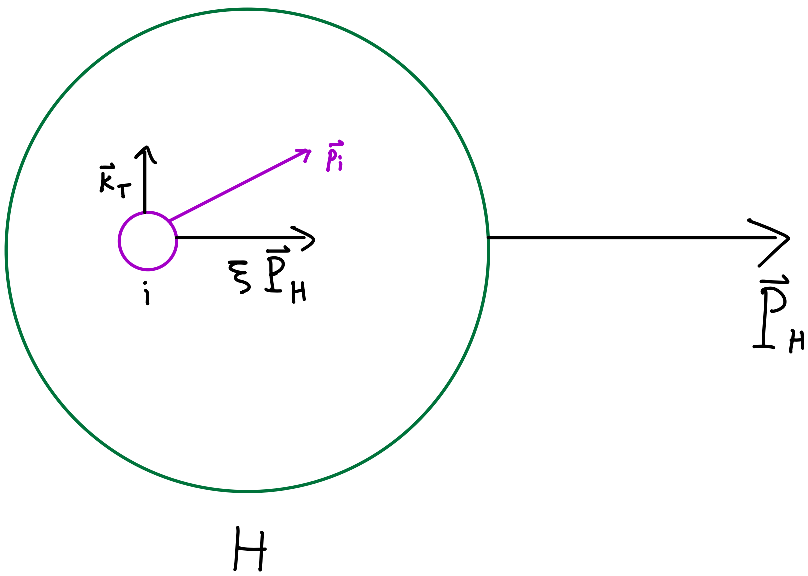
Drell-Yan process:

$pp \to \ell^+\ell^- + X$



Parton model:

$\vec{p}_i = \xi \vec{P}_H + \vec{k}_T$



Basic intuition: probability

$$P(A \cap B) = P(A) \times P(B|A)$$

probability that partons i, j of longitudinal fraction ξ_a, ξ_b exist and scatter

probability that partons i, j of longitudinal fraction ξ_a, ξ_b exist

$\sim f_{i/H_a}(\xi_a) f_{j/H_b}(\xi_b)$

probability that partons i, j of longitudinal fraction ξ_a, ξ_b scatter

$\sim d\hat{\sigma}_{ij}(\xi_a, \xi_b)$

→ sum over all possible i, j, ξ_a, ξ_b to get full Drell-Yan cross section

$$\frac{d\sigma}{dQ^2 dY} = \sum_{i,j} \int_{x_a}^{1} d\xi_a \int_{x_b}^{1} d\xi_b \; f_{i/H_a}(\xi_a) f_{j/H_b}(\xi_b) \frac{d\hat{\sigma}_{ij}(\xi_a, \xi_b)}{dQ^2 dY} \left[1 + \mathcal{O}\left(\frac{\Lambda_{QCD}^2}{Q^2} \right) \right]$$

kinematic variables:

$$Q^2 = (p_{\ell^+} + p_{\ell^-})^2 \; ; \; Y = \text{rapidity of } \ell^+\ell^- \; ; \; s = (P_{H_a} + P_{H_b})^2$$

$$x_a = Q e^{Y}/\sqrt{s} \; ; \; x_b = Q e^{-Y}/\sqrt{s}$$

Transverse momentum of $\ell^+\ell^-$ has been averaged over, so k_T of partons not important

If we want to measure $\vec{q}_T$ but it is large ($\sim Q \gg \Lambda_{QCD}$)

$$\frac{d\sigma}{d^4 q} = \sum_{i,j} \int_{x_a}^{1} d\xi_a \int_{x_b}^{1} d\xi_b \; f_{i/H_a}(\xi_a) f_{j/H_b}(\xi_b) \frac{d\hat{\sigma}_{ij}(\xi_a, \xi_b)}{dq} \left[1 + \mathcal{O}\left(\frac{\Lambda_{QCD}^2}{Q^2}, \frac{\Lambda_{QCD}^2}{q_T^2} \right) \right]$$

↳ now entire $\ell^+\ell^-$ momentum is measured

What about regime $\Lambda_{QCD} \lesssim q_T \ll Q$? 3 key things happen:

- transverse momenta of partons are now relevant
- longitudinal fractions ξ are fixed
- at LO parton types restricted to quarks/antiquarks of the same flavor

$$\frac{d\sigma}{d^4 q} = \frac{1}{s} \sum_{i \in \text{flavors}} \hat{\sigma}_{i\bar{i}}^{TMD}(Q) \int d^2 \vec{k}_T \; f_{i/H_a}(x_a, \vec{k}_T) \, f_{\bar{i}/H_b}(x_b, \vec{q}_T - \vec{k}_T) \left[1 + \mathcal{O}\left(\frac{q_T^2}{Q^2}, \frac{\Lambda_{QCD}^2}{Q^2} \right) \right]$$

$$= \frac{1}{s} \sum_{i \in \text{flavors}} \hat{\sigma}_{i\bar{i}}^{TMD}(Q) \int \frac{d^2 \vec{b}_T}{(2\pi)^2} e^{i \vec{b}_T \cdot \vec{q}_T} \tilde{f}_{i/H_a}(x_a, \vec{b}_T) \, \tilde{f}_{\bar{i}/H_b}(x_b, \vec{b}_T) \left[1 + \mathcal{O}\left(\frac{q_T^2}{Q^2}, \frac{\Lambda_{QCD}^2}{Q^2} \right) \right]$$

Focus on Drell-Yan $H_a = H_b = p$.

All of this has been classical intuition… incorporating QFT gives new dependencies from renormalization / regularization.

$\to$ only singular $1/\vec{q}_T^{\,2}$ pieces

$$\frac{d\sigma^W}{dQ\,dY\,d^2\vec{q}_T} = \sum_{i\in\text{flavors}} H_{i\bar{i}}(Q^2,\mu)\int d^2\vec{b}_T\, e^{i\vec{b}_T\cdot\vec{q}_T}\, \tilde{f}_{i/p}(x_a,\vec{b}_T,\mu,\zeta_a)\, \tilde{f}_{\bar{i}/p}(x_b,\vec{b}_T,\mu,\zeta_b)\left[1+\mathcal{O}\left(\frac{q_T^2}{Q^2},\frac{\Lambda_{QCD}^2}{Q^2}\right)\right]$$

↳ factorized: virtual corrections to hard process $q\bar{q}\to\gamma^*/Z\to\ell^+\ell^-$

What do these constituents look like in QFT?

$\to$ energetic radiation close to each proton

$$\tilde{f}_{i/p}(x,\vec{b}_T,\mu,\zeta) = \lim_{\substack{\epsilon\to 0\\ \tau\to 0}} Z^i_{uv}(\mu,\zeta,\epsilon)\,\frac{\tilde{f}^{0(u)}_{i/p}(x,\vec{b}_T,\epsilon,\tau,xP^+)}{\tilde{S}^{0\,subt}_{n_a n_b}(b_T,\epsilon,\tau)}\sqrt{\tilde{S}^0_{n_a n_b}(b_T,\epsilon,\tau)}$$

removes double counting

soft exchange between two partons

Z^i_{uv}: subtract UV divergences in $d=4-2\epsilon$ dimensions; introduces $\overline{MS}$ renormalization scale μ

τ: from rapidity divergence regulator; gives rise to Collins-Soper scale ζ (analagous to μ)

$\bar{\psi}\; W\; W\; W\; \psi$

$$\tilde{f}^{0(u)}_{i/p}(x,\vec{b}_T,\epsilon,\tau,xP^+)$$

$$= \int \frac{db^-}{2\pi}\, e^{-ib^-(xp^+)}\left\langle p(p)\right|\left[\bar{\Psi}^0_i(b^\mu)W_\sqsubset(b^\mu,0)\frac{\gamma^+}{2}\Psi^0_i(0)\right]_\tau\left|p(p)\right\rangle$$

$$\tilde{S}^0_{n_a n_b}(b_T,\epsilon,\tau) = \frac{1}{N_c}\left\langle 0\right|\mathrm{Tr}\left[W_\gtrdot(b_T)\right]_\tau\left|0\right\rangle$$

Explore these with an example.

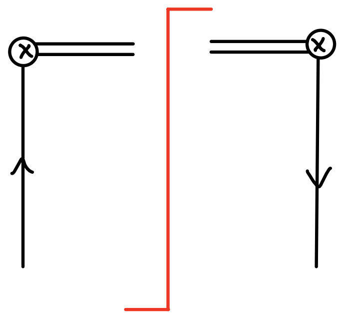
One-loop calculation

unsubtracted TMD PDF:

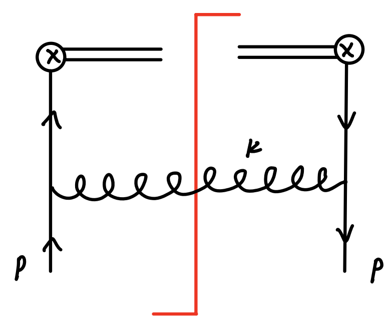
$$\tilde{f}^{0(u)}_{q/q'}(x, \vec{b}_T, \epsilon, \tau) =$$

$$\int \frac{db^-}{2\pi} e^{-ib^-(xp^+)} \langle q'(p) | \left[\bar{\psi}^0_q(b^\mu) W_\sqsubset(b^\mu, 0) \frac{\gamma^+}{2} \psi^0_q(0) \right]_\tau | q'(p) \rangle$$

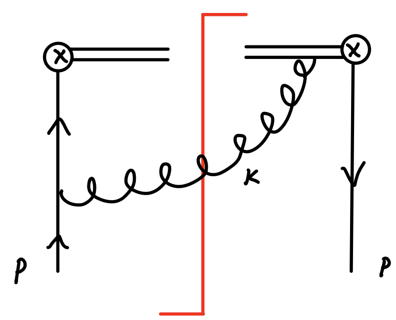
LO:



NLO: attach gluon in all possible ways. Only nonvanishing diagrams in dim reg are



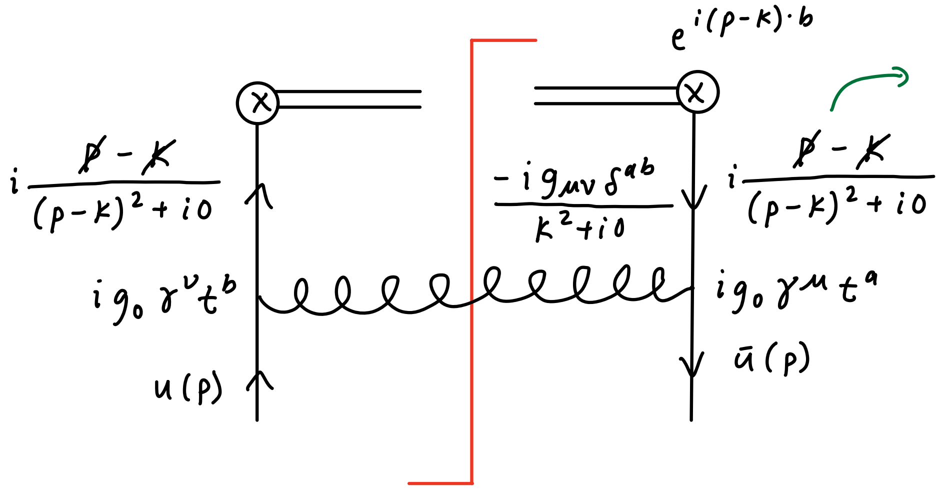
(a)



(b)

+ mirror

Look at first of these in detail.



$p = (p^+, 0, 0)$

$$\mathcal{M}_a = \int \frac{d^d k}{(2\pi)^d} \int \frac{db^-}{2\pi} e^{-ib^-(xp^+)}\, \bar{u}(p)(ig_0 \gamma^\mu t^a) \left[i\, \frac{\not{p} - \not{k}}{(p-k)^2 + i0} \right] e^{i(p-k)\cdot b}\, \frac{\gamma^+}{2}$$

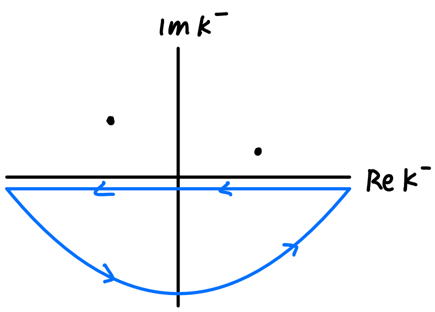
$$\times \left[i\, \frac{\not{p} - \not{k}}{(p-k)^2 + i0} \right] (ig_0 \gamma^\nu t^b)\, u(p) \left[\frac{-i g_{\mu\nu} \delta^{ab}}{k^2 + i0} \right]$$

$$\boxed{\int \frac{db^-}{2\pi} e^{-ib^-(xp^-)} e^{i(p-k)\cdot b} = \delta[(1-x)p^+ - k^+]\, e^{i\vec{b}_T \cdot \vec{k}_T}}$$

$$\mathcal{M}_a = i g_0^2 C_F \int \frac{d^{d-2}\vec{k}_T}{(2\pi)^d} e^{i\vec{b}_T\cdot\vec{k}_T} \int dk^- \frac{(2-d)(1-x)p^+}{[(p-k)^2 + i0](k^2 + i0)}$$

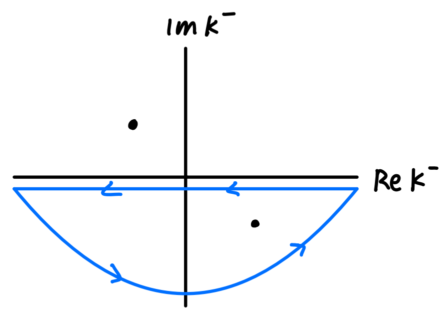
$$\boxed{\begin{aligned}
(p-k)^2 &= -2p^+k^- - 2\cancelto{0}{p^- k^+} + 2\cancelto{0}{\vec{k}_T\cdot\vec{p}_T} + \cancelto{0}{p^2} + k^2 \\
&= -2p^+k^- + 2k^+k^- - \vec{k}_T^2 \\
&= -2p^+k^- + 2k^-(1-x)p^+ - \vec{k}_T^2 \qquad (\delta \text{ function}) \\
&= -2xp^+k^- - \vec{k}_T^2 \\
k^2 &= 2k^+k^- - \vec{k}_T^2 \\
&= 2k^-(1-x)p^+ - \vec{k}_T^2
\end{aligned}}$$

Pole location: two k^- poles are on same side of real axis unless $0 < x < 1$:



$\to 0$

$x \geq 1$ or $x \leq 0$



$\neq 0$

$0 < x < 1$

$$\to \mathcal{M}_a = -\pi g_0^2 C_F (2-d)(1-x)\,\theta(x)\,\theta(1-x) \int \frac{d^{d-2}\vec{k}_T}{(2\pi)^d} e^{i\vec{b}_T\cdot\vec{k}_T} \frac{1}{\vec{k}_T^2}$$

Similar process for $\mathcal{M}_b$ ...

$d = 4 - 2\epsilon$

$$m_a + m_b = \frac{g_0^2 C_F}{2\pi}\left[\frac{1+x^2}{1-x} - \epsilon(1-x)\right]\int \frac{d^{d-2}\vec{k}_T}{(2\pi)^{d-2}}\,\frac{e^{i\vec{b}_T\cdot\vec{k}_T}}{\vec{k}_T^2}$$

renormalized coupling

$$= \frac{\alpha_s(\mu) C_F}{2\pi}\left[\frac{1+x^2}{1-x} - \epsilon(1-x)\right]\Gamma(-\epsilon)\left(\frac{\vec{b}_T^2\mu^2}{4e^{-\gamma_E}}\right)^{\epsilon}$$

$\hookrightarrow$ $x=1$: rapidity divergence!

Regulate by multiplying

$$R_c(k,\tau) = \left|\frac{\sqrt{2}\,k^+}{\nu}\right|^{-\tau} = \left[\frac{(1-x)p^+}{\nu/\sqrt{2}}\right]^{-\tau}$$

(equiv. to modifying Wilson line formulae)

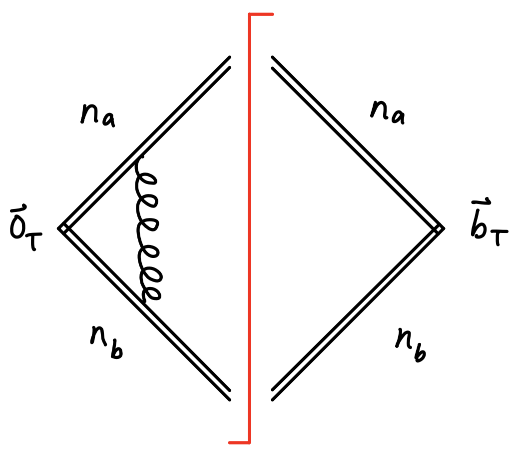
$$\rightarrow \frac{1+x^2}{1-x}(1-x)^{-\tau} = -\left(\frac{2}{\tau}+\frac{3}{2}\right)\delta(1-x) + \left[\frac{1+x^2}{1-x}\right]_+ + \mathcal{O}(\tau)$$

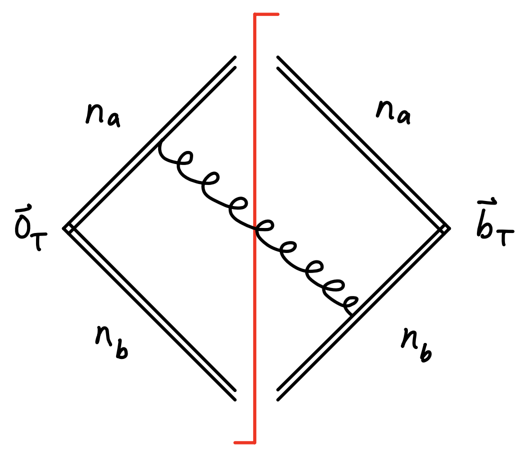
s.t. $[f(x)]_+ = f(x)$ for $x \neq 1$, $\int_0^1 dx\,[f(x)]_+ = 0$

So at one loop:

$$\tilde{f}_{q/q}^{\,0(u)}(x,\vec{b}_T,\epsilon,\tau) = \frac{\alpha_s(\mu)}{2\pi}C_F\left\{-\left(\frac{1}{\epsilon}+\ln\frac{\vec{b}_T^2\mu^2}{4e^{-2\gamma_E}}\right)\left[\frac{1+x^2}{1-x}\right]_+ + (1-x)\right.$$
$$\left. + \,\delta(1-x)\left(\frac{1}{\epsilon}+\ln\frac{\vec{b}_T^2\mu^2}{4e^{-2\gamma_E}}\right)\left(\frac{3}{2}+\frac{2}{\tau}-2\ln\frac{x p^+}{\nu/\sqrt{2}}\right) + \mathcal{O}(\tau,\epsilon)\right\}$$

Still need soft diagrams.





Evaluate using same methods.

$$\mathcal{M}_s = \frac{g_0^2 C_F}{4\pi} \int \frac{d^{2-2\epsilon} \vec{k}_T}{(2\pi)^{d-2}} \frac{e^{i\vec{k}_T \cdot \vec{b}_T}}{\vec{k}_T^2} \int_0^\infty \frac{dk^-}{k^-}$$

Regulate rapidity divergence:

$$\int_0^\infty \frac{dk^-}{k^-} \to \left(\frac{\nu}{\sqrt{2}}\right)^\tau \int_0^\infty \frac{dk^-}{k^-} \left| \frac{\vec{k}_T^2}{2k^-} - k^- \right|^{-\tau} = \frac{\nu^\tau k_T^{-\tau}}{2^\tau \sqrt{\pi}} \Gamma\left(\frac{1}{2} - \frac{\tau}{2}\right) \Gamma\left(\frac{\tau}{2}\right)$$

Soft function is

$$\tilde{S}_q^0(\vec{b}_T, \epsilon, \tau) = \frac{\alpha_s(\mu) C_F}{2\pi} \left[\frac{2}{\epsilon^2} + 4\left(\frac{1}{\epsilon} + \ln \frac{\vec{b}_T^2 \mu^2}{4e^{-2\gamma_E}}\right)\left(\frac{-1}{\tau} + \ln \frac{\mu}{\nu}\right) - \ln^2 \frac{\vec{b}_T^2 \mu^2}{4e^{-2\gamma_E}} - \frac{\pi^2}{6} \right] + \mathcal{O}(\tau, \epsilon)$$

In summary:

choose to cancel UV poles

$$\tilde{f}_{q/q}(x, \vec{b}_T, \mu, \zeta) = \lim_{\substack{\epsilon \to 0 \\ \tau \to 0}} Z_{uv}^i(\mu, \zeta, \epsilon) \, \tilde{f}_{q/q}^{0(u)}(x, \vec{b}_T, \epsilon, \tau, xP^+) \sqrt{\tilde{S}_q^0(b_T, \epsilon, \tau)}$$

subtraction factor $= 1$ with this regulation scheme

Final result:

IR pole

$$\tilde{f}_{q/q}(x, \vec{b}_T, \mu, \zeta) = \frac{\alpha_s(\mu) C_F}{2\pi} \left[-\left(\frac{1}{\epsilon} + \ln \frac{\vec{b}_T^2 \mu^2}{4e^{-2\gamma_E}}\right)\left[\frac{1+x^2}{1-x}\right]_+ + (1-x) - \frac{1}{2} \ln^2 \frac{\vec{b}_T^2 \mu^2}{4e^{-2\gamma_E}} + \ln \frac{\vec{b}_T^2 \mu^2}{4e^{-2\gamma_E}} \left(\frac{3}{2} + \ln \frac{\mu^2}{\zeta}\right) - \frac{\pi^2}{12} \right]$$

This result is independent of choice of rapidity regulator.